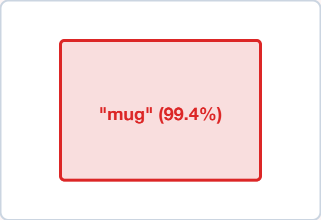
SPATIAL AFFORDANCE TOKENIZATION VS 2D CLASSIFICATION

Extracting SE(3) Contact Affordances & Metric 3D Bounding Boxes for Downstream Kinematic Execution

× 2D PASSIVE SEMANTIC CLASSIFICATION

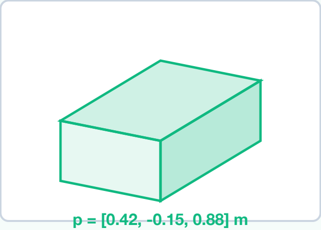
Disembodied Web Vision (e.g. ImageNet, CLIP)



- Zero metric depth (no distance z in meters)
- No contact normal vectors n̂ for grasping
- Scale ambiguity (toy mug vs industrial vessel)
- Blind to robot end-effector kinematics

✓ 3D SPATIAL AFFORDANCE TOKENIZATION

Metric Embodied Representation (e.g. DINOv2 3D / PointNet)



- Calibrated SE(3) centroid in world frame
- Estimated surface normals & friction cone
- Bounding collision primitive for CBF safety filter
- Metric affordance lease for trajectory planner